Data Validation

<u>Objective:</u>

Ensure that the synthetic dataset is logically sound, clean, and matches real-world expectations based on defined business rules and data dictionary specifications.

Validation Type	Description	Result
Null Checks	Checked for unexpected nulls in key fields (e.g., customer_id, booking_date, package_name)	✅ Passed
Data Type Checks	Ensured correct data types for all fields (e.g., age is integer, price_per_pax is float)	✅ Passed
Value Ranges	Verified ranges (e.g., age between 18–70, satisfaction_score between 1–10)	✅ Passed
Date Logic	Confirmed booking_date < travel_date, and refund_date falls between booking/travel	✅ Passed
Duplicate Check	Ensured no duplicate booking_id, customer_id, package_id in main tables	✅ Passed
Referential Integrity	All foreign key relationships (e.g., customer, package, guide IDs) resolve correctly	✅ Passed
Business Logic Rules	Validated: only refunded records have refund_status, only future travel = pending payment	✅ Passed
Skew Distribution	Verified skew in distributions (e.g., majority customers aged 25–40, higher ratings, low refund %)	✅ Passed

Data Validation Sample Codes:

```
# Null check
df.isnull().sum()

# Age range
assert df['age'].between(18, 70).all()

# Booking vs travel date
assert (df['booking_date'] < df['travel_date']).all()

# Refund logic
assert df[df['refund_status'].notnull()]['payment_status'].eq('refunded').all()
```